

Townsend 4.1-4.3

Questions for Gallicchio

I don't quite understand spin precession. I get that time evolution introduces a phase difference, and that the operator looked the same as the rotation operators, but intuitively this feels wrong / I don't know how we justify it.

4.1 The Hamiltonian and the Schrödinger Equation

Time evolution operator $\hat{U}(t)|\psi(0)\rangle = |\psi(t)\rangle$

infinitesimal time translation

$$\hat{U}(dt) = 1 - \frac{i}{\hbar} \hat{H} dt$$

$$\hat{U}(t) = \lim_{N \rightarrow \infty} \left[1 - \frac{i}{\hbar} \hat{H} \left(\frac{t}{N} \right) \right]^N = e^{-i \frac{\hat{H} t}{\hbar}}$$

4.2

The Schrödinger equation derived

4.3

Muons are ejected during some interaction opposite to spin, so we have a way of determining the spin direction for some systems. Something something sinusoids?